INTERSECTION NUMBERS WITH CURVES AGREE WITH DEGREES OF RESTRICTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's single mechanism sentence; the two definitional identifications it relies on are named in the span.

Allowed standard prerequisites (editorial).

- The leading term of a numerical polynomial has degree the dimension of the support. delivered companion lemma-numerical-polynomial-leading-term. Degree 1 of n -> chi(Z, L|_Z^{tensor n}).
- Definition of the intersection number. chapter definition definition-intersection-number. (L dot Z) as the leading coefficient.
- Definition of the degree of an invertible sheaf. chapter definition definition-degree-invertible-sheaf. deg(L|_Z) as the same consecutive difference; also the L-degree.

Source reference. Original definition 0AYR: sources/stacks-chow/tags.txt, source lines 13799–13799. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $Z \subset X$ be a closed subscheme of dimension ≤ 1. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Then*

$$(\mathcal{L} \cdot Z) = \deg(\mathcal{L}|_Z)$$

where $\deg(\mathcal{L}|_Z)$ *is as in Definition 0AYR. If $\mathcal{L}$ is ample, then* $\deg_{\mathcal{L}}(Z) = \deg(\mathcal{L}|_Z)$.

Proof. This follows from the fact that the function $n \mapsto \chi(Z, \mathcal{L}|_Z^{\otimes n})$ has degree 1 and hence the leading coefficient is the difference of consecutive values. □

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